

Ontology Documentation Summary

Ontology Scope

The ontology is designed to model autonomous driving corner-case scenarios by providing a structured and semantically rich representation of driving environments, entities, and associated safety risks. Its primary goal is to support consistent interpretation of complex traffic situations and enable reliable evaluation of Large Language Model (LLM) outputs.

It captures interactions between dynamic actors (such as vehicles and pedestrians), static infrastructure (such as lane markings and traffic signals), and environmental conditions (such as weather and visibility). In addition, it explicitly models potential safety risks arising from these interactions, allowing risks to be represented in a standardized and machine-readable format.

Key Classes

The ontology is organized into a hierarchical structure of classes that reflect real-world driving scenarios:

- Scene – Represents the overall driving environment (e.g., highway, intersection, urban road)
- Actor – Represents dynamic entities
 - Vehicle and Pedestrian are subclasses of Actor
- RoadObject – Represents static infrastructure such as lane markings, traffic signs, and obstacles
- TrafficSignal and TrafficSign – Represent traffic control elements
- WeatherCondition – Captures environmental context (e.g., rain, fog, night)
- EvidenceIndicator – Represents indirect cues indicating potential hazards
- Risk – Represents safety hazards in the scene

Each of these classes is enriched with `rdfs:label` and `rdfs:comment` annotations, which provide human-readable names and descriptions. These annotations improve interpretability and are also used in prompt generation to guide the model.

Risk Subclasses and Hierarchy

The **Risk** class is further divided into multiple subclasses to represent different types of hazards. These include:

- CollisionRisk
- ReducedVisibilityRisk

- LaneIntrusionRisk
- PedestrianCollisionRisk
- OcclusionRisk
- LowFrictionRisk
- SuddenObstacleRisk
- TrafficViolationRisk
- UnpredictableBehaviorRisk

In addition to this, the ontology also includes subclass relationships within broader concepts, meaning that risks and other entities are part of a structured hierarchy

All risk subclasses are also annotated with labels and comments, ensuring that each risk type has a clear definition. This helps both in human understanding and in guiding the model during ontology-based prompting.

Key Properties

Object Properties

Object properties define relationships between entities:

- hasActor – Links a scene to actors
- hasVehicle / hasPedestrian – Specify actor types
- hasRoadObject / hasObstacle – Link scene to static elements
- hasWeatherCondition – Captures environmental context
- hasRisk – Associates risks with the scene
- occursInScene – Links risks back to the scene

These relationships allow the ontology to represent interactions between entities in a structured way.

Datatype Properties

Datatype properties store descriptive and numerical information:

- hasSeverityScore – Numerical risk severity (0–1)
- hasDescription – Textual explanation of the risk
- hasScenarioName – Identifies the scenario

Major Modeling Decisions

Several design decisions were made to ensure clarity and usability:

- A hierarchical class structure was used to reflect real-world relationships
- Risks were modeled as explicit ontology classes instead of free text
- Classes include labels and comments to improve interpretability and guide LLM prompting
- Environmental context was separated into dedicated classes
- Relationships were kept simple and interpretable for easier parsing and evaluation

Extensions Beyond the Paper

More Detailed Risk Categories

Additional subclasses such as SuddenObstacleRisk, UnpredictableBehaviorRisk, and LowFrictionRisk were introduced to capture more nuanced hazards.

Additional Environmental Context

Classes such as NightCondition and FoggyCondition were added to better represent real-world driving conditions.

Evidence Indicators

Classes like RollingBallIndicator and OccludedObjectIndicator were introduced to model indirect cues of potential risks.